

Predicting Heart Disease Risk Using Machine Learning: An Exploratory and Comparative Analysis

Big Data Technologies for the degree
of
MSc
in
Advanced Computer Science with
Software Engineering



Department of
Computer and Information Sciences

November 2025

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Chapter 1

Introduction

Cardiovascular diseases (CVDs) remain the primary cause of death in the world, accounting for an estimated 19.8 million deaths each year (World Health Organisation, 2023). Early diagnosis of potential patients at risk of heart disease is therefore vital for suitable intervention and prevention. Advances in data science and machine learning have enabled the development of predictive models that can analyse clinical and behavioural data to support medical decision-making.

The purpose of this study is to investigate the clinical and behavioural factors associated with heart disease and to develop a predictive model for heart failure risk using machine learning methods. By analysing the Heart Failure Prediction Dataset, which includes key clinical features such as age, blood pressure, cholesterol, fasting blood sugar, and exercise angina, this research aims to identify patterns that distinguish healthy individuals from those with heart disease.

The analysis begins with data cleaning and exploratory data analysis to understand the distribution and relationships of features. Summary statistics and visualisations are used to identify potential trends and outliers. Next, an unsupervised learning approach (K-Means clustering) is applied to explore whether patients naturally group based on their clinical characteristics. Finally, a supervised learning model (Logistic Regression) is used to predict the presence of heart disease and evaluate its accuracy using standard performance metrics such as accuracy, F1-score, and ROC-AUC.

By combining exploratory, unsupervised, and supervised analyses, this study demonstrates how data analysis methods can be applied to real healthcare datasets to enhance understanding of disease risk factors and support predictive modelling for improved clinical outcomes.

Chapter 2

Exploratory Data Analysis

1 Aims

Investigate clinical and behavioural factors associated with heart disease and develop a predictive model for heart failure risk using machine learning techniques.

2 Source

The dataset used is publicly available on Kaggle (Fedesoriano, 2021). It was created by previously merging several independent datasets together. In this dataset, five heart disease datasets were merged based on 11 shared features, resulting in the largest heart disease dataset currently available for research. The datasets integrated include Cleveland (303 records), Hungarian (294 records), Switzerland (123 records), Long Beach VA (200 records), and Stalog (Heart) Data Set (270 records), giving a total of 1190 observations. After removing 272 duplicate entries, the final dataset comprises 918 unique observations.

3 Analysis

Boxplots for all numeric variables were generated as seen in **Error! Reference source not found.** to assess data distribution and detect outliers (Age, RestingBP (Resting Blood Pressure), Cholesterol, FastingBS (Fasting Blood Sugar), MaxHR (Maximum Heart Rate Achieved), Oldpeak, and HeartDisease). This visualisation allows for quick comparison of data ranges, median values and the presence of outliers across variables.

The distribution of age values appears fairly compact with no extreme outliers. The median age is around 50 years, and most patients fall between 40-60 years, suggesting that middle-aged people

make up the majority of the dataset. Resting Blood Pressure (RestingBP) is around 130 mmHg. A few outliers were perceived at the lower and higher ends, possibly due to data entry mistakes.

The Fasting Blood Sugar (Fasting BS) is a binary feature (0 or 1), so the boxplot does not provide much valuable insight. However, most values appear at 0, indicating that most patients have normal fasting blood sugar levels.

Cholesterol shows a wide range, with values lying between 180 and 270 mg/dl. The boxplot depicts several outliers at about 500 mg/dl, which could represent patients with hypercholesterolemia (abnormally high cholesterol) (American Heart Association, 2024). While some other outliers with a value of 0 mg/dl were detected at the lower end. A 0-cholesterol value does not make sense biologically; it might be a data entry error. These outliers should be removed as they may mislead the model and bias the mean, standard deviation and correlation values.

The value range of Maximum Heart Rate (MaxHR) lies between 70 and 200 bpm, with a median near 140 bpm. While a few lower outliers suggest patients with limited cardiac capacity. Oldpeak values are concentrated near zero, with a few higher outliers (approximately up to 5). Meaning that most patients don't show much ST depression.

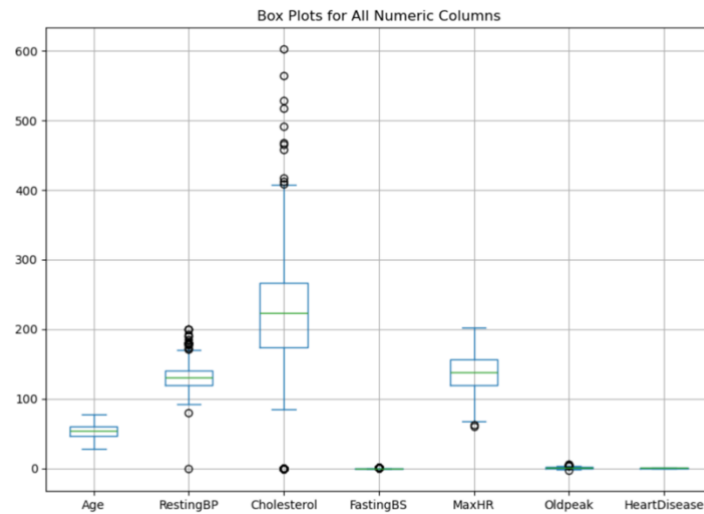


Figure 1: Box Plot for all Numeric Columns

Figure 2 shows the boxplot after removing the null cholesterol values. It is noticeable that new outliers appeared in the 80-100 mg/dL range; this occurred because the previously included value of 0 mg/dL was far below the physiological range (normal cholesterol levels are typically at around 100 mg/dL (American Heart Association, 2024), which had stretched the boxplot's y-axis and masked these moderate outliers. As a result, values in the 80-100 range looked visually merged near the zero baseline. They were hidden visually due to scaling. After removing the 0 values, the boxplot automatically rescaled its y-axis. The lower outliers in RestingBP and MaxHR had disappeared as well, which proves that the outliers were likely originated from an entry mistake.

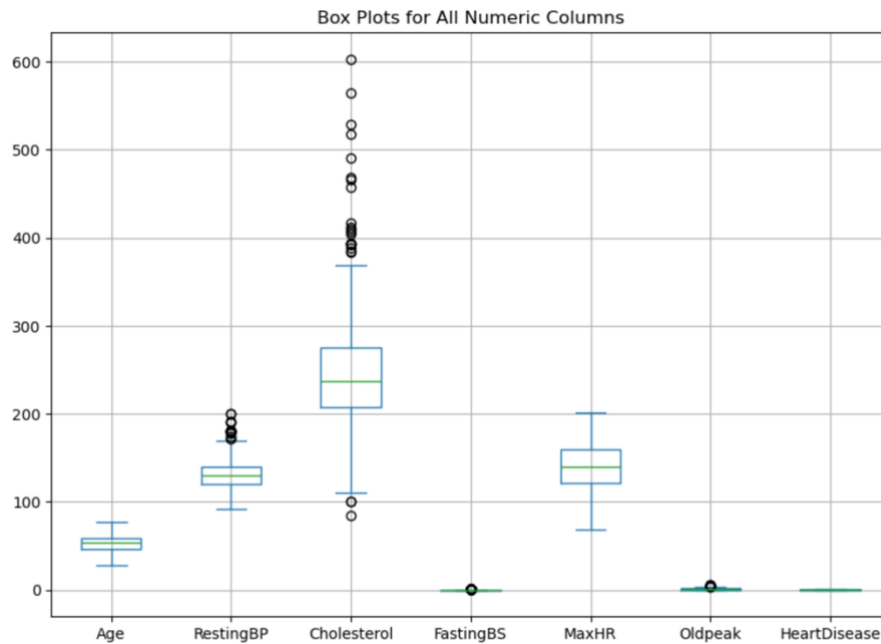


Figure 2: Boxplots after removing outliers.

A correlation heatmap was generated (Figure 3) to identify relationships between the variables. The heatmap shows that none of the variables exhibit strong correlations ($|r| \geq 0.5$) with heart disease. The results indicate that Oldpeak ($r = 0.40$) and MaxHR ($r = -0.40$) show the strongest correlations with HeartDisease, which suggests that increased ST depression and reduced maximum heart rate are key indicators for heart problems, while Age and FastingBS also show weak positive associations. This means that heart disease is influenced by a combination of factors, rather than a single dominant factor.

Cholesterol and RestingBP display weak correlations, indicating limited predictive value on their own. This suggests that while these features are clinically relevant, their impact could depend on other stronger factors.

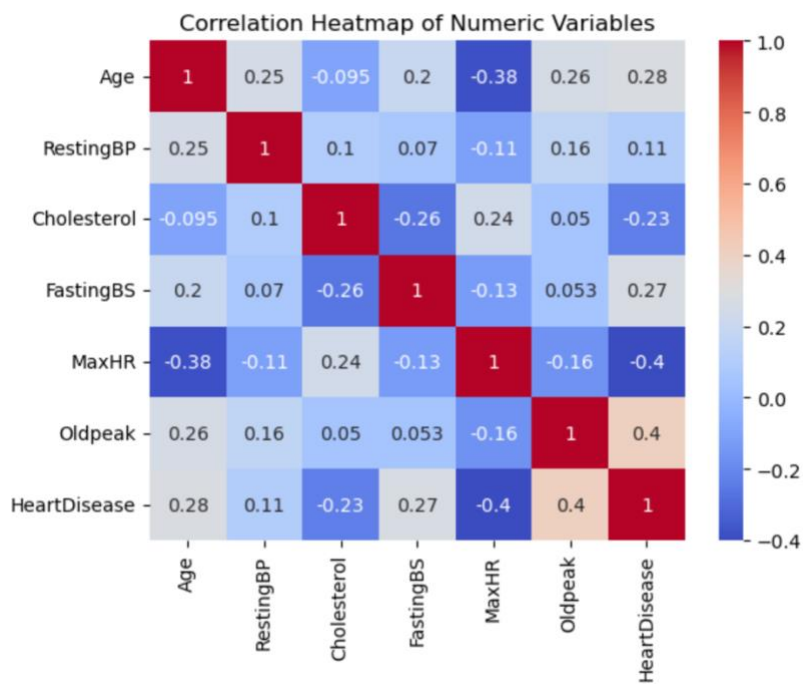


Figure 3: Correlation Heatmap of Numeric variables

Chapter 3

Unsupervised Analysis – Clustering

For the unsupervised analysis, K-Means was selected as it is one of the most effective methods for numerical datasets. The method efficiently partitions data into distinct groups based on feature similarity, which is suitable for this study's aim of exploring hidden patterns among patients with varying heart disease risks. Given that most of the features in the dataset are numerical, K-Means is an appropriate choice for this study.

String (categorical) features such as "Sex" and "chestPain" were excluded because K-Means relies on distance calculation, which cannot be calculated using non-numeric variables.

The variable FastingBS was retained in the clustering analysis despite being binary, as it provides clinically valuable information regarding elevated blood sugar levels. Although K-Means treats this as a continuous variable, standardisation ensured that it contributed proportionally to the distance calculations without dominating other features.

The variable Sex was excluded from clustering because, although binary, it represents a nominal category without a continuous numerical interpretation. Including it could bias cluster distances, as K-Means would treat gender differences as quantitative rather than categorical. In contrast, FastingBS was retained, as its binary values carry meaningful clinical significance (normal vs elevated fasting blood sugar)

To ensure fair distance comparison, all numeric features were standardised using z-score normalisation. Before scaling, the features exhibited varying magnitudes (e.g., cholesterol around 245 mg/dl versus FastingBS around 0.17). After standardisation, each variable achieved a mean of 0 and a standard deviation of 1, which ensures a fair contribution to clustering algorithms.

The completeness and homogeneity scores were low (≈ 0.1 – 0.2), indicating that the clusters generated by K-Means did not align strongly with the true heart disease labels. This suggests that the patient data does not separate cleanly into distinct risk groups. This outcome is consistent with the

multifactorial nature of heart disease, where overlapping risk factors (e.g., age, cholesterol, and blood pressure) contribute in complex, non-linear ways to the overall condition.

To visualise the K-Means clustering results, a two-dimensional Principal Component Analysis (PCA) was performed. PCA is a dimensionality reduction technique that transforms correlated features into a smaller set of uncorrelated components, ordered by decreasing variance (Jolliffe and Cadima, 2016). In this analysis, PCA was used to visualise multidimensional clustering results in two dimensions, in order to facilitate the interpretation of the clustering structure.

PCA reduced the six standardised features into two principal components, capturing the majority of the dataset's variance. The resulting scatter plot (Figure 4) shows three clusters ($k = 3$) with noticeable overlap, indicating that patient profiles are not easily separable. This observation supports the low clustering metrics obtained earlier and suggests that heart disease risk factors interact in complex, continuous ways rather than forming discrete patient groups.

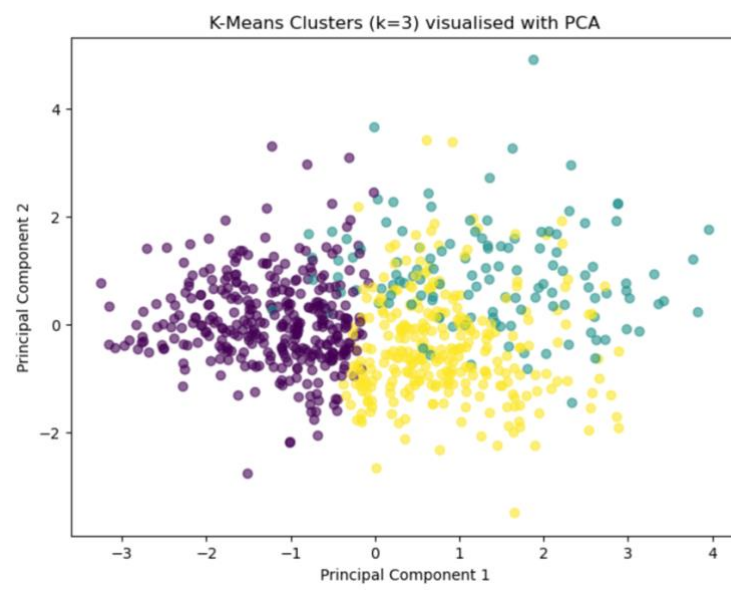


Figure 4: K-Means Clusters visualised with PCA

Chapter 4

Supervised Analysis

This chapter aims to develop a predictive model for heart failure risk using machine learning techniques. To achieve this, the likelihood of heart disease ($\text{HeartDisease} = 1$) will be predicted based on clinical attributes.

Machine learning models only understand numeric data. However, the dataset includes string/categorical features (Sex, ChestPainType, RestingECG, ExerciseAngina, St-slope). As they may carry valuable information that allows to create the predictive model. Thus, binary categorical features were encoded.

For the other features, to convert them into a numeric format that algorithms can understand, a method used in data preprocessing (especially in machine learning) called one-hot encoding is applied.

Logistic Regression was chosen for the supervised analysis because it is well-suited for binary classification problems. It can directly generate predictions that fall between 0 and 1. Moreover, it handles both continuous and categorical variables effectively after appropriate preprocessing (scaling and encoding). All numeric variables were standardised using StandardScaler to ensure compatibility of coefficients. Binary categorical variables (Sex and ExerciseAngina) were encoded as 0 and 1. While categorical variables with more than 3 options (ChestPainType, RestingECG and ST_Slope) were one-hot encoded using dummy variables.

The data was then divided into 70% training and 30% testing subsets using `train_test_split`.

The confusion matrix table (see **Error! Reference source not found.**) shows that the model correctly identified 243 out of 276 cases (80%). Only 33 misclassifications occurred (13 false positives and 20 false negatives).

	Predicted No	Predicted Yes
Actual No	99	13
Actual Yes	20	144

Table 1: Confusion Matrix

metrics	Heart Disease = 1	Heart Disease = 0
Precision	0.92	0.83
Recall	0.88	0.88
F1-score	0.9	0.86

Table 2: Table performance Logistic Regression method

A classification report was also generated. Precision measures how accurate the positive predictions are. A value of 0.92 (92%) was recorded, which is a high precision level, meaning that the model correctly identifies most actual patients.

Recall measure how well the model catches all real cases of heart disease. High recall means the model rarely misses true cases, which is very important in healthcare, where missing a positive case can be critical. The model achieved an F1-score of 0.90, indicating a strong balance between precision and recall. This suggests that the model performs well in both correctly identifying patients with heart disease and minimising false positives.

The ROC Curve (Receiver Operating Characteristic curve) is a graphical way to evaluate the performance of a classification model. It was implemented to evaluate how well the model distinguishes between patients with heart disease (1) and without it (0).

The x-axis represents False Positive Rate (FPR), which is the proportion of healthy patients incorrectly classified as having disease. While the y-axis is for True Positive Rate (TPR), which is also called Recall or Sensitivity. It shows how many actual heart disease cases the model correctly identified.

The Logistic Regression model achieved a ROC-AUC of 0.945, which means that there is a 94.5% that the model will rank sick patients higher than healthy ones. The curve's position near the top-left corner confirms strong sensitivity and specificity. This demonstrates the effectiveness of the model in distinguishing between at-risk and non-risk patients.

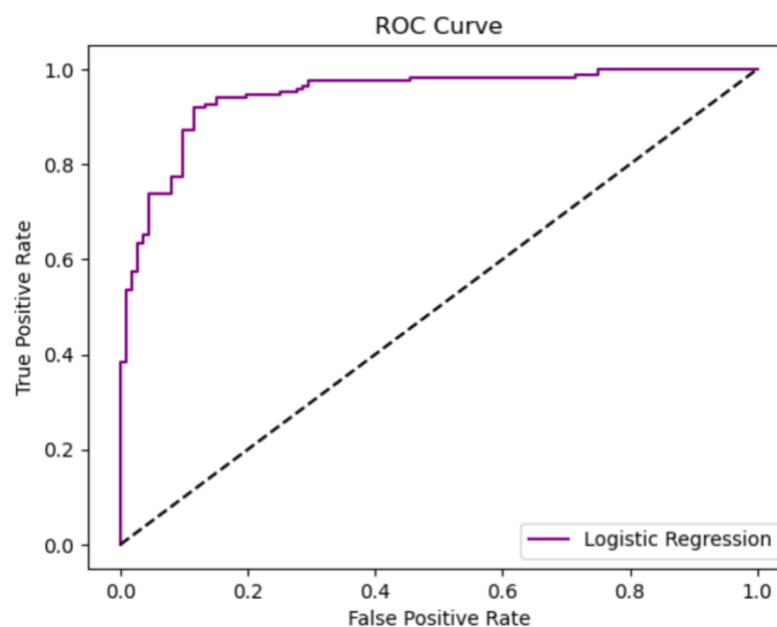


Figure 5: ROC Curve

To see which features contribute mostly to heart disease, the coefficients were transformed into odds ratios using the exponential function. Positive coefficients (odds ratio >1) indicate factors that increase the likelihood of heart disease, while negative coefficients (odds ratio < 1) correspond to protective factors (reduce the likelihood of heart disease). ST_Slope_Flat (2.06) and ExerciseAngina (1.64) were the strongest risk predictors, whereas MaxHR (0.96) and ChestPainType (0.57) reduced the likelihood of heart disease.

	Coefficient	Odds Ratio
ST_Slope_Flat	0.724129	2.062934
ExerciseAngina	0.495610	1.641499
Oldpeak	0.490402	1.632972
FastingBS	0.419695	1.521498
Age	0.156728	1.169677
RestingBP	0.023468	1.023745
MaxHR	-0.042836	0.958069
RestingECG_Normal	-0.057284	0.944326
RestingECG_ST	-0.155521	0.855969
ChestPainType_TA	-0.241235	0.785657
Cholesterol	-0.411979	0.662338
ST_Slope_Up	-0.460063	0.631244
Sex	-0.489955	0.612654
ChestPainType_ATA	-0.567316	0.567045
ChestPainType_NAP	-0.609886	0.543413

Figure 6: Coefficients and Odd Ratios of the features

Chapter 5

Discussion

The present analysis explored the Heart Failure Prediction Dataset to identify key clinical factors associated with heart disease and to develop a machine-learning model capable of predicting heart failure risk. Through data exploration and statistical analysis, several meaningful insights were derived about patient characteristics and their relationship with heart disease outcomes.

Exploratory analysis revealed that variables such as age, maximum heart rate (MaxHR), exercise-induced angina, ST segment slope, and fasting blood sugar exhibited clear differences between healthy individuals and those with heart disease. In particular, patients with heart disease tended to be older, had lower MaxHR, and were more likely to present with exercise-induced angina and ST-segment depression (Oldpeak). These observations are consistent with established medical knowledge, confirming the dataset's validity.

Unsupervised clustering analysis (K-Means) suggested that patient profiles could not be neatly partitioned into distinct clusters, as indicated by low completeness and homogeneity scores. This reinforces the multifactorial nature of heart disease, where risk factors interact in complex, overlapping ways.

The supervised learning phase employed Logistic Regression, which achieved strong predictive performance:

- Accuracy: 88%
- F1-score: 0.90
- ROC-AUC: 0.95

These results demonstrate that the model was effective at distinguishing between patients with and without heart disease. High F1 and ROC-AUC scores indicate a strong balance between sensitivity

(recall) and specificity (precision), which is critical in medical prediction tasks. The confusion matrix showed few false negatives, which is important in clinical contexts, as missing a potential heart disease case could have serious implications.

The logistic regression coefficients indicated that ST_Slope_Flat, ExerciseAngina, Oldpeak, and FastingBS were strong positive predictors of heart disease, meaning that their presence or increase raised the likelihood of heart failure. In contrast, higher MaxHR, normal resting ECG, and certain chest pain types (e.g., Typical Angina) acted as protective factors. These results align closely with medical literature, where ST-segment abnormalities and exercise-induced angina are established indicators of cardiovascular stress.

Chapter 6

Reflections

Conducting this study provided valuable experience in the whole process of data analysis, from selecting useful data to answer a research question, to understanding and cleaning data to apply machine learning models for prediction. One of the most significant lessons learned, was the importance of data preprocessing and exploration. Detecting and handling outliers and interpreting their meaning, missing values and categorical variables required careful attention, as these early steps directly influence the model's performance and reliability.

While the dataset performed well, several limitations must be acknowledged. The dataset, although useful, was relatively small (918 samples) and may not generalise to all populations. Furthermore, it didn't include certain features that may directly or indirectly contribute to heart disease, such as smoking, diet, which are important in heart disease prediction.

I also gained a deeper understanding of model evaluation metrics, especially precision, recall, F1-score, and ROC-AUC. These measures provided a more meaningful assessment than accuracy alone, particularly in a medical context where misclassifying patients can have serious consequences.

Although I initially encountered challenges with data encoding and scaling, overcoming these issues strengthened my understanding of how different variable types affect machine learning models. I also learned the importance of maintaining data integrity, for example, verifying unrealistic cholesterol values and understanding their impact on visualisations.

For future work, I might consider exploring more advanced supervised methods, such as decision trees, Random Forest, to capture potential non-linear relationships and interactions between clinical features that Logistic Regression may not fully represent.

Chapter 7

Conclusion

This study successfully achieved its aim of investigating clinical and behavioural factors associated with heart disease and developing a predictive model for heart failure risk using machine learning techniques. Several meaningful findings emerged through systematic data cleaning, exploratory analysis, and model evaluation.

The exploratory analysis revealed that features such as age, maximum heart rate (MaxHR), exercise angina, ST segment slope, FastingBS and Oldpeak were key indicators of heart disease. These variables showed clear differences between healthy and affected patients, consistent with established medical research.

The Logistic Regression model demonstrated strong predictive capability, achieving an accuracy of 88%, an F1-score of 0.90, and an ROC-AUC of 0.95. These results indicate a strong balance between precision and recall, confirming the model's effectiveness in identifying individuals at risk of heart disease.

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